RealQM as Plasma Physics?

A metal from Coulomb geometry, and the Fermi surface as the open question

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Abstract

In this article we extend RealQM — developed for atoms, molecules and nuclei, where matter is modelled as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy — to the plasma physics of **two-component systems**: ions and electrons, both charged and both free to move. A metal is such a system, as are a stellar interior and the recombination era of a Coulomb cosmology; and so, notably, is RealQM itself, which contains discrete charges and no smeared continuum anywhere. The central result is that RealQM's prescription for a metal *coincides* with the Wigner–Seitz construction: one electron domain per ion, filling the cell, with the normal derivative vanishing at the cell face by symmetry. For sodium it gives an equilibrium radius $r_s = 3.77\,a_0$ and bulk modulus 9.1 GPa, against a standard treatment's $3.87\,a_0$ and 8.0 GPa and measured values $3.93\,a_0$ and 6.3 GPa — agreement within the accuracy of the model, from a framework carrying neither an exchange term nor a band-filling term. The plasma frequency $\omega_p^2 = 4\pi n$ comes out exactly, since a rigid displacement leaves the localisation energy untouched. What the framework does *not* have is a Fermi surface, and that is the open question: de Haas–van Alphen oscillations, the linear electronic specific heat and positive plasmon dispersion are measured in real metals, not in any idealisation, and RealQM has neither bands nor E_F to account for them. A closing remark treats the uniform electron gas, where the localisation term vanishes identically — a result about a construction containing a smeared inert background, which RealQM does not contain, and so a diagnosis of mismatched ontology rather than a test the framework fails.

1 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to the plasma physics of **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system. So is a stellar interior, and so is the recombination era of a Coulomb cosmology.

The extension is natural in a way worth stating at the outset, because it is not true of every framework: a two-component plasma is what RealQM is already made of. The construction contains discrete unit charge densities on their own domains and no smeared continuum anywhere, so a system of ions and electrons is not an idealisation the theory must be adapted to describe — it is the theory's own furniture, in a new arrangement.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i on non-overlapping domains Ω_i, with

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j} \iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i} \psi_i^2 = 1$, the interfaces free to move to where the energy says. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 A metal is a Wigner–Seitz problem, and RealQM already agrees with it

Take a metal: a lattice of ions, one valence electron each. RealQM assigns each electron its own domain; by translational symmetry those domains are the Wigner–Seitz cells, one ion at the centre of each; and at a cell face, neighbouring cells being identical, the normal derivative of the density vanishes.

That is exactly the **Wigner–Seitz construction**, used for alkali metals since 1933. The framework does not improvise in condensed matter — it arrives already agreeing with an established method, and differs from it in one term only. Standard theory solves the same cell problem for the band bottom and then *adds* the mean kinetic energy of filling the band, $\frac{3}{5}E_F$, together with exchange and correlation, all consequences of the exclusion principle. RealQM keeps the electrostatics, takes its kinetic energy from the cell state itself, and adds nothing: non-overlap is supposed to do that work.

For sodium, with an Ashcroft empty-core pseudopotential ($r_c = 1.67\,a_0$), electrostatics $-0.9/r_s + 1.5\,r_c^2/r_s^3$ for a point ion in a neutral sphere, exchange $-0.458/r_s$ and Wigner correlation, minimising the energy per electron over r_s gives

	RealQM	standard	measured
equilibrium r_s (a_0)	3.77	3.87	3.93
bulk modulus (GPa)	9.1	8.0	6.3

Table 1: Sodium. The standard column reproduces the metal, which is what makes the comparison meaningful; RealQM sits beside it, within the accuracy of the model.

The framework thus gives the equation of state of a metal from Coulomb geometry alone, to within a few per cent on the lattice scale and within the model's own error on the stiffness, while carrying no exchange term and no band-filling term.

The mechanism of that agreement should be stated plainly rather than claimed as a triumph. At $r_s = 3.93$ the cell state carries $\langle T \rangle = 0.004$ Ha against a band-filling term $\frac{3}{5}E_F = 0.0715$ Ha, so RealQM is *not* reproducing degeneracy pressure. It arrives at a comparable answer because it also omits exchange, -0.117 Ha, a term of opposite sign and comparable size, so that their effects on the curvature largely cancel. That cancellation is the framework's own design claim — that non-overlap does the work exchange does — and here it is borne out numerically. It is not a demonstration that the two omitted terms are individually unnecessary.

3 The plasma frequency, exactly

Displace the electron distribution rigidly by ξ against the ion background. The resulting surface charge produces a restoring field $4\pi n\xi$, so $\ddot{\xi} = -4\pi n\xi$ and

$$\omega_p^2 = 4\pi n, \tag{2}$$

free of temperature and of any statistical assumption. RealQM reproduces this exactly, for a structural reason: under a rigid translation each ψ_i is carried along unchanged, so $\nabla\psi_i$ is unchanged, the localisation term contributes nothing, and what remains is Coulomb and inertia — which is all the result depends on.

The same structure suggests what the framework should do about **Landau damping**. RealQM is not a single fluid: N electrons are N domains, each carrying its own current and drift velocity, which is a multi-beam plasma. Multi-beam models reproduce Landau damping as phase mixing among cold streams, and do so reversibly, in agreement with plasma-echo experiments. Where the velocity spread is thermal, collisionless damping should therefore emerge for the right reason rather than by an inserted term. This has not been computed.

4 Conduction, where the framework is least settled

Electric conduction is the interaction of the electrons with the ion lattice, and here the position is genuinely uncertain.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$ for every material, making copper behave like a semiconductor. Whether it does has not been computed, and the pessimistic reading should be held loosely: the same lattice that carried the metal in §2 is present here too.

5 What remains open: the Fermi surface

The framework has no bands and no E_F, and there are measurements bearing on this directly. They are made on real two-component metals, not on any idealisation, so they cannot be set aside by choosing the object of study more carefully:

- **de Haas–van Alphen oscillations** map the Fermi surface of copper and aluminium directly; its existence and shape are not in doubt;
- the **electronic specific heat** is linear in temperature, with a coefficient fixed by the density of states at E_F;
- the **plasmon dispersion** is positive and measured, while the compressibility term that produces it, $\omega^2 = \omega_p^2 + \frac{3}{5}k^2 v_F^2$, is absent here.

No account of these is currently available in the framework, and constructing one is the substantial open problem. It is a sharper problem than it would have appeared a section ago: not that RealQM lacks Fermi pressure — Table 1 shows a metal's equation of state coming out regardless — but that a set of measured Fermi-surface phenomena have no representation in a theory built from localised domains.

Remark: the uniform electron gas

It is worth recording what happens if the framework is applied to the *uniform electron gas*, since the calculation is exact and that system is often taken as the reference problem.

Partitioning a gas of density n into non-overlapping cells of size $a = n^{-1/3}$ gives a localisation cost per cell scaling as a^{-2}, hence a kinetic energy density $t(n) = C\, n^{5/3}$ — the Thomas–Fermi form, obtained from the geometry of a partition rather than from Fermi statistics. Linearising (1) about the uniform state and substituting into Poisson's equation gives Yukawa screening with $k^2 = 18\pi n^{1/3}/5C$, and setting C to the Thomas–Fermi coefficient $C_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3} = 2.8712$ reproduces k_{TF} exactly at every density. The *form* extends without adjustment.

The coefficient does not. With the free interface used throughout RealQM, $\psi_i \equiv |\Omega_i|^{-1/2}$ is admissible: it satisfies the normalisation, makes the total density uniform, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Hence $C = 0$ exactly, the screening length is infinite and the plasmon dispersionless. Neither allowing the interfaces to move nor requiring domains to be connected and singly charged alters this.

This should not be read as a test the framework fails. The uniform electron gas is electrons plus a *smeared, inert, non-responding* positive background — a construction of a tradition in which exchange–correlation energies are fitted to that very system. RealQM contains no inert backgrounds anywhere, so asking it to reproduce jellium asks it to reproduce a system whose central ingredient it does not have, and $C = 0$ diagnoses that mismatch. Consistently with this, the pathology disappears as soon as real ions are present: Table 1 is the same framework on the same physics, with the background replaced by the charges matter is actually made of.

One physical case does approach the uniform limit closely: degenerate matter in which the ion lattice is a small perturbation, as in a white dwarf interior, where the measured mass–radius relation depends on precisely the pressure the uniform limit does not supply. That is the one place the result might have observable consequences, and it is untested here.

A note on what was tested numerically

Equation (2), the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. Table 1 is a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional; the standard column reproducing sodium is what makes the comparison meaningful, and the residual 27% error in its bulk modulus is the accuracy of that model rather than of the framework. An attempt to test the interface condition on helium is omitted: it was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion was drawn from it. Scripts accompany the source.

References

[1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, in [2].

[2] C. Johnson, *Real Quantum Mechanics* (Body&Soul, Vol. VI), `https://claes542.github.io/RealMolecule/realquantum3.pdf`.

[3] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, in [2].